Dhruvraj Singh Shekhawat

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Professional Summary

Electronics and Communication Engineering student at VIT Bhopal University (CGPA: 8.3/10) with hands-on experience delivering 3 production-style projects in Machine Learning, Computer Vision, Full-Stack Web Development, and MATLAB-based engineering simulations. Proficient in 4 programming languages and industry tools including Python, Java, TensorFlow, Scikit-learn, Flask, OpenCV, and MATLAB. Achieved a 5 star HackerRank rating with a strong foundation in software engineering and AI.

Education

VIT Bhopal University — Expected May 2027
B.Tech in Electronics and Communication Engineering — *CGPA: 8.3/10*

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Deep Learning, Signals & Systems, Digital Electronics, Data Visualization, Object-Oriented Programming

Technical Skills

Programming Languages	Python, Java, C, MATLAB
Machine Learning	TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas
Signal Processing	MFCC Feature Extraction, Audio Processing, Feature Engineering
Web Development	Flask, HTML, CSS, JavaScript, Bootstrap
Engineering Tools	MATLAB, Simulink, Git, Linux, Figma
Core Competencies	Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Signal Processing, Software Engineering, REST APIs,

Experience

Technical Team Member, FYI Club — Sep 2025 – Apr 2026

VIT Bhopal University

- Organized and supported 10+ technical workshops, coding events, and student engagement activities involving multidisciplinary teams.
- Maintained technical infrastructure, event logistics, and operational workflows to ensure smooth execution of club activities.
- Collaborated with student developers and organizers to streamline event planning and improve participant experience.
- Provided technical troubleshooting and deployment support during workshops and technology-focused events.
- Promoted programming, software development, and emerging technologies through peer learning initiatives.

Projects

Audio Spoof Detection System

Python · TensorFlow · Scikit-learn · MFCC · NumPy

- Developed an end-to-end audio spoof detection pipeline using TensorFlow and the AASIST architecture to classify genuine and spoofed speech.
- Preprocessed and extracted MFCC features from **8,000+ audio samples**, implementing normalization, feature engineering, and augmentation to improve model robustness.
- Designed and optimized training, validation, and testing workflows using TensorFlow and Scikit-learn, reducing preprocessing time through automated data pipelines.
- Achieved **98% classification accuracy** with strong Precision, Recall, and F1-score performance on unseen evaluation data.
- Detected replay, synthesized, and voice-converted speech attacks using deep learning and signal processing techniques.
- Built a modular, reusable pipeline supporting feature extraction, model training, evaluation, and inference for future experimentation.

GraminSeva – Rural Issue Reporting Platform

Flask · HTML · CSS · JavaScript · Bootstrap

- Implemented CRUD operations and complaint status tracking using Flask.
- Designed responsive interfaces compatible with desktop and mobile devices using Bootstrap.
- Structured backend modules for maintainability, scalability, and efficient data management.
- Integrated frontend and backend components for seamless issue reporting.

Dynamic Wireless Power Transfer Analysis

MATLAB · Simulink

- Simulated DD, DDQ, Circular, and Rectangular coil topologies using MATLAB and Simulink.
- Analyzed coupling coefficient variation, coil misalignment, and power transfer efficiency.
- Compared 4 coil geometries to identify configurations with improved alignment tolerance and efficiency.
- Performed simulation-driven engineering analysis for dynamic EV wireless charging systems.
- Documented comparative performance evaluation and visualization results.

Certifications

- Google IT Support Professional Certificate — Google
- Applied Machine Learning in Python — University of Michigan (Coursera)
- Introduction to IoT — NPTEL
- Introduction to Programming in C — NPTEL
- Python Essentials — VIT Bhopal